

[ACJC/2022/Q1]

- 1 Wordle is a popular word game where players are made to guess a five-letter word within six tries. The task is to recreate the game in Python.

Task 1.1

Write a function `get_word(filename)` that:

- accepts `filename`, a string representing the name of a file containing one or more five letter words;
- randomly chooses a word from the file and returns the word. [3]

Test the function with `words.txt`.

For example,

```
get_word('words.txt')
```

should return a word randomly chosen from `words.txt`. [1]

Task 1.2

Write a function `check_validity(guess)` to check if guess is valid.

A valid guess should:

- contain only lowercase letters;
- not contain numbers, punctuation, spaces or any other symbols;
- be exactly five letters in length.

The function would return `True` if guess is valid and `False` if not. [3]

Test the function fully with suitable test cases. [2]

Task 1.3

Write a function `check_guess(guess, word)` that:

- accepts two parameters:
 - o `guess`: a string that represents what the player guessed;
 - o `word`: a string that represents the answer;
- creates two lists:
 - o the list `correct` is a list of indices of letters in `guess` that are in `word` and in the correct position;
 - o the list `in_word` is a list of indices of letters in `guess` that are in `word` but not in the correct position;
- returns a list containing `correct` and `in_word` [6]

Test the function with the following test cases:

`check_guess('maple', 'apple')` should return `[[2, 3, 4], [1]]`.

`check_guess('poppy', 'apple')` should return `[[2], [0]]`.

`check_guess('apple', 'apple')` should return `[[0, 1, 2, 3, 4], []]`.

`check_guess('apepe', 'apple')` should return `[[0, 1, 4], [3]]`. [3]

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Task 1.4

Write a function `display_result(correct,in_word)` to inform the player of the results of his guess.

The function:

- accepts two parameters, the lists `correct` and `in_word`;
- outputs a string containing only the following characters: `^*_`, where
 - `^` represents a letter that is in the word and in the right position;
 - `*` represents a letter that is in the word but not in the right position;
 - `_` represents a letter that is not in the word;

[4]

The output of the function should be displayed to the player.

Test the function with the following test cases:

`display_result([2,3,4],[1])` should display `'_***'`.

`display_result([0,1,2,3,4],[1])` should display `'^*****'`.

`display_result([], [1])` should display `'_*____'`.

[3]

Task 1.5

The function `play_wordle()` for the game can be found in `Task1_5.txt`. The function `play_wordle()` makes use of the functions you created in **Tasks 1.1 to 1.4** to run the game.

Copy the `play_wordle()` function from `Task1_5.txt` into your Jupyter Notebook.

Run the program.

[1]

Download your program code and output for Task 1 as

`TASK1_<your name>_<centre number>_<index number>.ipynb`